

A Appendix

A.1 INLP Guarantees

In this section, we prove, for the binary case, an orthogonality property for INLP classifiers: each two classifiers w_i and w_j from two iterations steps i and j are orthogonal (Lemma A.1). Several useful properties of the matrix P that is returned from INLP emerge as a direct result of orthogonality: the product of the projection matrices calculated in the different INLP steps is commutative (Corollary A.1.1); P is a valid projection (Corollary A.1.2); and P projects to a subspace which is the intersection of the nullspaces of all INLP classifiers $N(w_1) \cap N(w_2) \cap \dots \cap N(w_n)$ (Corollary A.1.3). Furthermore, we bound the influence of P on the structure of the representation space, demonstrating that its impact is limited only to those parts of the vectors that encode the protected attribute (Lemma A.2).

We prove those properties for two consecutive projection matrices P_1 and P_2 from two consecutive iterations of Algorithm 1, presented below in 5. The general property follows by induction.

1. $w_1 = \operatorname{argmin}_w \mathcal{L}(w; X; Z)$
2. $P_1 := P_{N(w_1)} = \text{GetProjectionMatrix}(N(w_1))$
3. $X' = P_1 X$
4. $w_2 = \operatorname{argmin}_w \mathcal{L}(w; X' = P_1 X; Z)$
5. $P_2 := P_{N(w_2)} = \text{GetProjectionMatrix}(N(w_2))$

INLP Projects to the Intersection of Nullspaces.

Lemma A.1. *if w_2 is initialized as the zero vector and trained with SGD, and the loss $\mathcal{L}$ is convex, then w_2 is orthogonal to w_1 , that is, $w_1 \cdot w_2 = 0$.*

Proof. In line 4 of the algorithm, we calculate $w_2 = \operatorname{argmin}_w \mathcal{L}(w; X' = P_1 X; Z)$. For a convex $\mathcal{L}$ and a linear model w , it holds that the gradient with respect to w is a linear function of x : $\nabla_w \mathcal{L}(x_i) = \alpha_i x_i$ for some scalar α_i . It follows that after t stochastic SGD steps, w_2^t is a linear combination of input vectors $x_1, \dots, x_i, \dots, x_t$. Since we constrain the optimization to $x_i \in N(w_1)$, and considering that fact the nullspace is closed under addition, at each step t in the optimization it holds

that $w_2^t \in N(w_1)$. In particular, this also holds for the optimal w_2^* . $\square$

We proceed to prove commutativity based on this property.

Corollary A.1.1. $P_1 P_2 = P_2 P_1$

Proof. By Lemma A.1, $w_1 \cdot w_2 = 0$, so $P_{R(w_1)} P_{R(w_2)} = P_{R(w_2)} P_{R(w_1)} = 0$, where $P_{R(w_i)}$ is the projection matrix on the row-space of w_i . We rely on the relation $P_{N(w_i)} = I - P_{R(w_i)}$ and write:

$$P_1 P_2 = (I - P_{R(w_1)})(I - P_{R(w_2)}) = I - P_{R(w_1)} - P_{R(w_2)} - P_{R(w_1)} P_{R(w_2)} = I - P_{R(w_1)} - P_{R(w_2)}.$$

Similarly,

$$P_2 P_1 = (I - P_{R(w_2)})(I - P_{R(w_1)}) = I - P_{R(w_2)} - P_{R(w_1)} - P_{R(w_2)} P_{R(w_1)} = I - P_{R(w_1)} - P_{R(w_2)}, \text{ which completes the proof. } \square$$

Corollary A.1.2. $P = P_2 P_1$ is a projection, that is, $P^2 = P$.

Proof. $P^2 = (P_2 P_1)^2 = P_2 P_1 P_2 P_1 \stackrel{*}{=} P_2 P_2 P_1 P_1 = P_2^2 P_1^2 \stackrel{**}{=} P_2 P_1 = P$, where $*$ follows from Corollary A.1.1 and $**$ follows from P_1 and P_2 being projections. $\square$

Corollary A.1.3. $P_2 P_1$ is a projection onto $N(w_1) \cap N(w_2)$.

Proof. Let $x \in \mathbb{R}^n$. $P_2(P_1 x) \in N(w_2)$, as P_2 is the projection matrix to $N(w_2)$. Similarly, $P_2(P_1 x) = P_1(P_2 x) \in N(w_1)$, so $Px \in N(w_1) \cap N(w_2)$. Conversely, let $x \in N(w_1) \cap N(w_2)$. Then $P_1 x = x = P_2 x$, so $Px = P_2 P_1 x = P_2 x = x$, so x is mapped by P to $N(w_1) \cap N(w_2)$. $\square$

Note that in practice, we enforce Corollary A.1.3 by using the projection Equation 1 (section 4). As such, the matrix P that is returned from Algorithm 1 is a valid projection matrix to the intersection of the nullspaces even if the conditions in Lemma A.1 do not hold, e.g. when $\mathcal{L}$ is nonconvex or w_2 is not initialized as the zero vector.

⁷If we performed proper dimensionality reduction at stage 3 – i.e., not only zeroing some directions, but completely removing them – the optimization in 4 would have a unique solution, as the input would not be rank-deficient. Then, we could use an alternative construction that relies on the Representer theorem, which allows expressing w_2 as a weighted sum of the inputs: $w_2 = \sum_{x_i \in X' = P_1 X} \alpha_i x_i$, for some scalars α_i . As each x_i is inside the nullspace, so is any linear combinations of them, and in particular w_2 .

INLP Approximately Preserves Distances.

While the projection operations removes the protected information from the representations, ostensibly it could have had a detrimental impact on the structure of the representations space: as a trivial example, the zero matrix $\mathbf{O}$ is another operator that removes the protected information, but at a price of collapsing the entire space into the zero vector. The following lemma demonstrate this is not the case. The projection minimally damages the structure of the representation space, as measured by distances between arbitrary vectors: the change in (squared) distance between $x, x' \in \mathbb{R}^n$ is bounded by the difference between the “gender components” of x and x' .

Lemma A.2. *Let $\vec{w} \in \mathbb{R}^n$ be a unit gender direction found in one INLP iteration, and let $x, x' \in \mathbb{R}^n$ be arbitrary input vectors. Let $P := P_{N(\vec{w})}$ be the nullspace projection matrix corresponding to $\vec{w}$. Let $d(x, x') := \|x - x'\|$ and $d^P(x, x') := \|Px - Px'\|$ be the distances between x, x' before and after the projection, respectively. Then the following holds:*

$$(d(x, x') - d^P(x, x'))^2 \leq (x\vec{w} - x'\vec{w})^2$$

Proof. notation: we denote the i th entry of a vector x by x_i .

Since $\vec{w}$ is the parameter vector of a gender classifier, a point $x \in \mathbb{R}^n$ can be classified to a gender $g \in \{0, 1\}$ according to the sign of the dot product $x\vec{w}$. Note that in the binary case, the nullspace projection matrix P is given by

$$P = I - \vec{w}\vec{w}^T \quad (5)$$

Where $\vec{w}\vec{w}^T$ is the outer product. By definition, if $\vec{w}$ is in the direction of one of the axes, say without loss of generality the first axis, such that $\vec{w} = [1, 0, \dots, 0]$, then the following holds:

$$\vec{w}\vec{w}^T = \begin{bmatrix} 1 & 0 & \dots & 0 \\ \vdots & 0 & & \\ & & \ddots & \\ 0 & \dots & & 0 \end{bmatrix} \quad (6)$$

Such that $\vec{w}\vec{w}^T$ is the zero matrix except its $(1, 1)$ entry, and then P is simplified to

$$P = \begin{bmatrix} 0 & 0 & \dots & 0 \\ \vdots & 1 & 0 \dots & 0 \\ 0 & & \ddots & \\ 0 & \dots & & 1 \end{bmatrix} \quad (7)$$

I.e, the unit matrix, except of a zero in the $(1, 1)$ position. Hence, the projection operator Px keeps x intact, apart from zeroing the first coordinate x_1 . We will take advantage of this property, and rotate the axes such that $\vec{w}$ is the direction of the first axis. We will show that the results we derive this way still apply to the original axes system.

Let R be a rotation matrix, such that after the rotation, the first coordinate of Rx is aligned with $\vec{w}$:

$$(Rx)_1 = x\vec{w} \quad (8)$$

One can always find such rotation of the axes. Let $x' \in \mathbb{R}^n$ be another point in the same space. Given the original squared distance between x and x' :

$$d(x, x') = \|x - x'\|^2 \quad (9)$$

Our goal is to bound the squared distance between the projected points in the new coordinate system:

$$d^{P,R}(x, x') := \|[P]_R Rx - [P]_R Rx'\|^2 \quad (10)$$

Where $[P]_R$ denotes the projection matrix P in the rotated coordinate system, which takes the form 7.

Note that R , being a rotation matrix, is orthogonal. By a known result in linear algebra, multiplication by orthogonal matrices preserves dot product and distances. That means that the distance is the same before and after the rotation: $d^{P,R}(x, x') = d^P(x, x')$, so we can safely bound $d^{P,R}(x, x')$ and the same bound would hold in the original coordinate system.

By 7,

$$\begin{aligned}
d^{P,R}(x, x') &= \sqrt{(0-0)^2 + \sum_{i=2}^n ([Rx]_i - [Rx']_i)^2} \quad (11) \\
&= \sqrt{d(x, x')^2 - ([Rx]_1 - [Rx']_1)^2}
\end{aligned}$$

Note that in general it holds that for any $a \geq b \geq 0$

$$\begin{aligned}
\sqrt{a-b} &= \sqrt{a+b-2b} = \sqrt{a+b-2\sqrt{b}\sqrt{b}} \\
&\geq \sqrt{a+b-2\sqrt{a}\sqrt{b}} = \sqrt{(\sqrt{a}-\sqrt{b})^2} \\
&= \sqrt{a}-\sqrt{b}
\end{aligned} \quad (12)$$

Combining 12 with 11 when taking $a = d(x, x')^2, b = ([Rx]_1 - [Rx']_1)^2$ we get:

$$d^{P,R}(x, x') \geq d(x, x') - |([Rx]_1 - [Rx']_1)| \quad (13)$$

From 11 one can also trivially get

$$\begin{aligned}
d^{P,R}(x, x') &= \sqrt{d(x, x')^2 - ([Rx]_1 - [Rx']_1)^2} \\
&\leq \sqrt{d(x, x')^2} = d(x, x')
\end{aligned} \quad (14)$$

Combining 14 and 13 we finally get:

$$\begin{aligned}
d(x, x') - |([Rx]_1 - [Rx']_1)| &\leq d^{P,R}(x, x') \\
&< d(x, x')
\end{aligned} \quad (15)$$

Or, equivalently, after subtracting $d(x, x')$ from all elements and multiplying by -1:

$$\begin{aligned}
|([Rx]_1 - [Rx']_1)| &= |x\vec{w} - x'\vec{w}| \\
&\geq d(x, x') - d^{P,R}(x, x') \geq 0
\end{aligned}$$

So

$$\begin{aligned}
(d(x, x') - d^{P,R}(x, x'))^2 & \\
&\leq ([Rx]_1 - [Rx']_1)^2 = (x\vec{w} - x'\vec{w})^2
\end{aligned}$$

Note that this result has a clear interpretation: the difference between the distance of the projected x, x' and the distance of the original x, x' is bounded by the difference of x and x' in the gender direction $\vec{w}$. In particular, if x and x' are equally male-biased, their distance would not change at all; if x is very male-biased and x' is very female-biased, the projection would significantly alter the distance between them.

□